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Session : Technical Paper Presentation Session

An Integrated System for Water Distribution Network Management 21630

Presented By

Moreshwar Salpekar, Architect – Systems Engineering, Sevya Multimedia Pvt Ltd

INTRODUCTION

- Large Scale Water Networks
- Underground pipes
- Recurring problems such as leakage, bursts, pressure transients, and contamination.
- Not so easy to detect and repair
- Preventive Maintenance work will help
- Real-Time Data based prioritization and problem solving

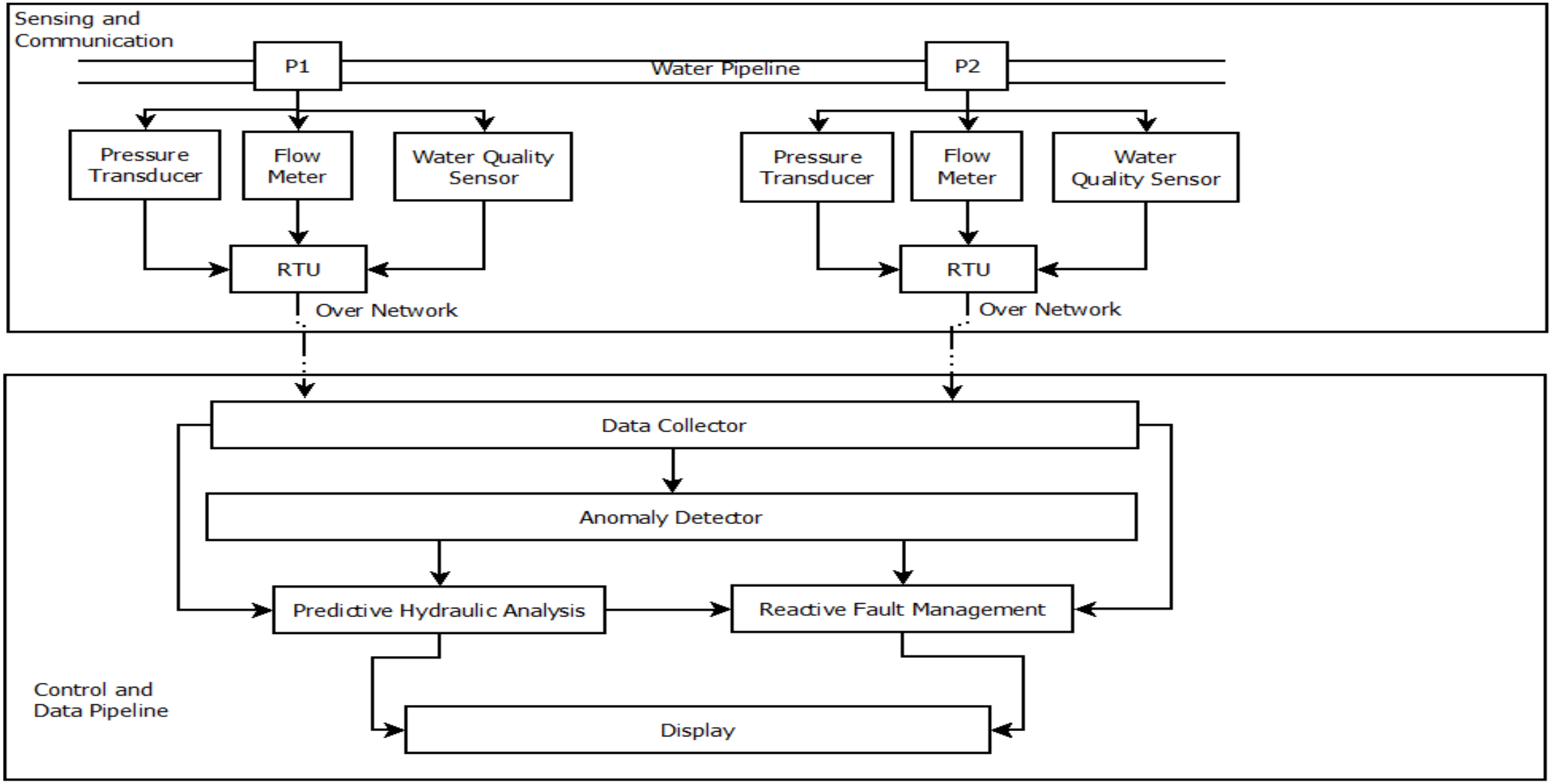
CONTEXT

- Reactive maintenance: contamination scenario, no water scenario, high costs.
- Predictive maintenance : Possibly big candidate lists requiring prioritization and need data from all of pipeline network
- Prior work “Smart Water Distribution Using Low Cost Communication Protocol”:
  - Secure, low-cost IoT sensing and an ML classifier for anomaly detection
  - Did not integrate rapid source localization. Probabilistic maintenance prioritization pipeline wasn’t added

RELEVANCE

- Engineers
  - Know the system and can fix the problems
  - But can’t track each point on network – Need Help
  - Sensors can be added collect data and send via Remote Telemetry Units (RTUs) at different points in the network
    - Even with real-time data, lot of effort to isolate and prioritize proactive and reactive maintenance
- Machine learning systems /Embedded Systems to the aid
  - Aid decision making
  - Work on collected data
    - Process data
    - Identify and prioritize Proactive and Reactive maintenance points

The Proposed System



Reactive + Predictive Risk Prioritization Example

Segment	ΔPressure (bar)	Flow ↑ (%)	ML Burst Flags (30d)	Age (yrs)
S1 – Residential	0.4	12%	1	12
S2 – Near Hospital	1.5	45%	4	22
S3 – Industrial	1.0	30%	2	30
Assumed design life: 40 years				

Feature	Weight
Pressure (a)	1.4
Flow (b)	1.2
ML Frequency(c)	1.1
Age (d)	0.8
Bias (e)	-1.5

Priority Formula (P_i)
z_i = a * p_feature_i + b * q_feature_i + c * ml_freq_i + d * age_i + e
Pi = 1/(1+ exp(- z_i))

Normalized Values				
Segment	p	q	ml	age
S1	0.20	0.24	0.10	0.30
S2	0.75	0.90	0.80	0.55
S3	0.50	0.60	0.40	0.75

Rank	Segment	Failure Probability	Action
1	S2	0.74	Immediate inspection
2	S3	0.65	Planned intervention
3	S1	0.32	Monitor

ΔP = P_baseline-P_current, Flow Increase % = ((Q_current-Q_baseline)/Q_baseline) ×100
ML Burst Flags = Number of anomaly events flagged by the ML model in the last 30 days for that segment, Age = Time since installation or last full pipe replacement.
Pressure Normalization – (ΔP /2), Flow normalization – (Flow Increase %) /50 ML Normalization - ML Flags /5 Age Normalization – Age/40
Bias of -1.5 -> when everything except b is 0, failure probability is (1/(1+e^1.5)) = 1/5.48 = 18.2% which is reasonable. However, bias = -1.5 + aging_factor * fault_age_months. E.g. Open anomaly 3 months: aging_penalty = 0.2 * 3 = 0.6 effective_bias = -0.9. aging_penalty can prevent p_feature and q_feature from becoming 1.

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)